

Coding Tests

Welcome! As part of the interview process, please complete **one** of the three coding tests below. Choose the test that best matches the role you're applying for and where you feel most confident.

Option 1: Full-Stack Engineering

Build a Product Catalog API and Frontend

You'll build a REST API with Express.js + SQLite and a React frontend for an e-commerce product catalog. The backend handles CRUD operations, search, filtering, and pagination. The frontend displays a product listing with search and a detail page.

Time	1–2 days
Stack	Node.js, Express.js, SQLite, React, Vite
Deliverables	7 API endpoints (products, categories, stats) + 2 React pages
Best for	Candidates interested in web development and full-stack roles

Option 2: Data Science & Analytics

Analyze Brazilian E-Commerce Data and Forecast Demand

You'll work with ~100K real orders from a Brazilian marketplace (2016–2018). Explore the data, compute business metrics, build a demand forecast for the next quarter, and use an LLM to synthesize your findings into actionable recommendations.

Time	2–3 hours
Stack	Python, pandas, Jupyter, forecasting libraries, OpenRouter API
Deliverables	5 analysis functions + Jupyter notebook with EDA and visualizations
Best for	Candidates interested in data science, analytics, and ML

Option 3: AI Infrastructure

Build a Customer Support Workflow with LangGraph

You'll design and implement an AI-powered support workflow using LangGraph. The system classifies customer messages by intent, routes them through appropriate handling logic, generates responses using an LLM, and decides whether to escalate to a human agent.

Time	2–3 hours
Stack	Python, LangGraph, OpenRouter API
Deliverables	4 workflow functions: graph architecture, intent classification, escalation logic, end-to-end processing
Best for	Candidates interested in AI engineering, LLM applications, and agent systems

Instructions

1. **Pick one test** — candidates should pick their own test and you only need to complete one.
2. Follow the README.md inside the test directory for setup instructions and detailed requirements.
3. Each test includes smoke tests you can run to verify your solution before submitting.
4. Submit your completed work by providing either a repository link or a zip file.

What We Evaluate

Across all tests, we look for:

- **Correctness** — Does the solution work and meet the requirements?
- **Code quality** — Is the code clean, readable, and well-organized?
- **Reasoning** — Are decisions explained and justified?
- **Communication** — Is the work easy to follow and review?

Good luck!